Universität Konstanz

Abstraction Mechanisms for Virtual Avatars

Master's Thesis Defense

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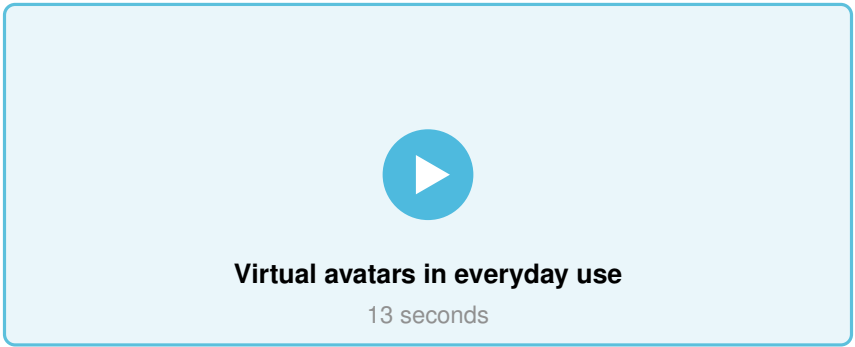
Second reviewer: Prof. Dr. Tiare Feuchtner

University of Konstanz, 2026

Motivation



Virtual avatars in everyday use

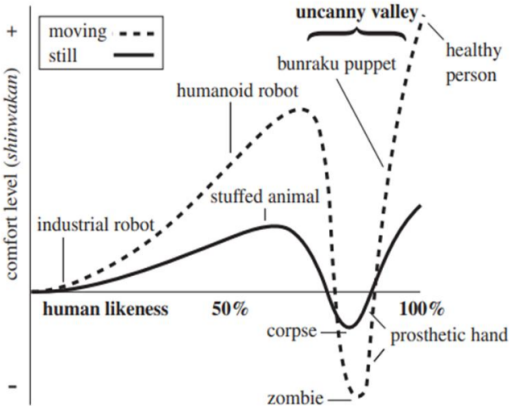
13 seconds

They present, they teach, they advise and they sell. How they are drawn decides how much we believe them.

Clips: Lex Fridman Podcast #398 · Avaturn · Apple Vision Pro Spatial Persona (YouTube). Full credits at the end.

Problem Statement

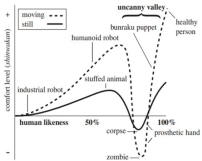
- Comfort rises with **human likeness**, but only up to a point
- That dip: **the uncanny valley**, proposed by Mori for humanoid robots [1]
- Later confirmed for **CG characters**

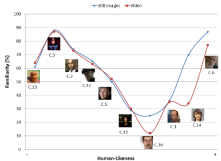


Related Work

1. Measuring the valley

From a drawn curve, to data, to a validated instrument, to CG characters





01

1970, Mori

The Uncanny Valley [1]

Proposed it. Comfort drops just short of human, and **motion deepens the dip**.

02

2006, MacDorman & Ishiguro

The Uncanny Advantage of Using Androids [2]

Measured it. Eeriness peaks where **comfort is lowest**.

03

2010, Ho & MacDorman

Revisiting the Uncanny Valley Theory [3]

Instrumented it. Eeriness becomes **separable from likeness**.

04

2012, Dill et al.

Evaluation of the Uncanny Valley in CG Characters [4]

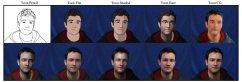
Extended it. The same curve appears for **CG characters**.

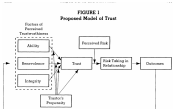
What this gives the thesis: a valley that is real, measurable and known to reach CG avatars. **What it does not give:** any evidence about whether a viewer would *rely* on the character.

2. Trust follows the same curve

How far the valley reaches, and how trust is measured

Why trust and not comfort: an advisor people do not believe is useless, however comfortable it is to look at. The question is whether the valley reaches that far.





01

2005, Nowak & Rauh

The Influence of the Avatar on Online Perceptions [5]

Tested it on avatars. More human-like avatars were rated more credible, and chosen more often.

02

2012, McDonnell et al.

Render Me Real? [6]

Measured trust directly. Across eleven render styles it comes out valley-shaped.

03

2020, Natarajan & Gombolay

Effects of Anthropomorphism and Accountability [7]

Beyond avatars too. Anthropomorphic agents were trusted more.

04

So how is trust measured?

Mayer et al., An Integrative Model of Organizational Trust, 1995 [8]

Ability, benevolence, integrity. Three separable dimensions. This study measures all three.

Measuring Trust

Trustworthiness is a property of the trustee, and it has three parts

Ability

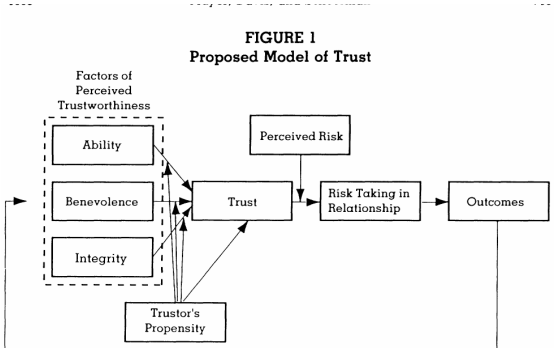
Competence inside one domain, not in general.

Benevolence

The belief that the other party cares about my wellbeing.

Integrity

Sticking to principles I accept, and saying what they mean.



Why three and not one: they move independently. Someone can be entirely competent and still not act in your interest, and one score would hide which part moved [8].

3. Is abstraction the exit?

Four studies, pointing in three directions

01

2024, Canales et al.

The Impact of Avatar Stylization on Trust [9]

Measured all three dimensions plus a forced choice. Self-report flat, **behaviour not**. Identity beat style.

02

2025, Dubosc et al.

Effect of Avatar Stylization and Facial Expression Intensity [10]

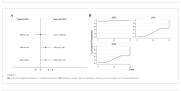
Interacted face to face in VR. Stylised avatars were rated **significantly less eerie**.

03

2025, Gao et al.

Trust in Virtual Agents [11]

Set an agent against the real speaker. The **real speaker** scored higher on all three dimensions.



04

2025, Tao et al.

Impact of Virtual Avatar Appearance Realism [12]

Pooled thirteen studies. High realism was rated **more trustworthy** than low.

And the verdict: Alipour et al. reviewed 53 studies and found the link **inconsistent** [13]. Nobody has tested one avatar under both treatments against the real person it was built from.

Research questions

RQ1 Technical transfer

How can NPR stylisation techniques be adapted to avatar platforms with different rendering architectures, and what capabilities and limitations does each platform impose?

RQ2 Technique selection

Which NPR technique or combination of techniques best balances visual abstraction against facial and expressive legibility under production rendering constraints?

The system

RQ3 Perception

How does NPR stylisation affect perceived eeriness and trustworthiness, and can an NPR avatar reach the trustworthiness of a photorealistic avatar and of a real human speaker?

The user study

The first two are answered by what was built. The third by a study with 41 participants, where this talk spends most of its time.

Methodology & Evaluation

Three avatar platforms

Chosen to span the range from open pipeline to closed interface

Mixamo

Meta Avatars SDK

Avaturn

	Mixamo	**Meta SDK**	**Avaturn**
Shader access	Full URP, unrestricted	**Hook injection only**, after SDK lighting	Full URP, unrestricted
Facial animation	**None**	Requires Quest Pro hardware	**Blendshape animation**
Key challenge	No facial rigging	Shader manipulation	Initial Unity import
Strength	Full shader freedom	Native Quest 3 integration	**High fidelity facial quality**

The plan: build the same set on all three and see what survives the move. Mixamo for prototyping, the Meta SDK for live tracked animation in VR, and Avaturn for the study, the only one combining open shader access with a face rebuilt from photographs.

Eight techniques

Assessed against four criteria on the Avaturn avatar

Technique	Facial readability	Temporal	Artefact suppression	Expressive abstraction	Cand.
V1 Outline [14]	✓	✓	(✓) alpha cutout	× not in isolation	✓
V2 Toon [15]	✓	✓	✓	(✓) no interior lines	—
V2.2 X-Toon [16]	✓	✓	✓	(✓) manual mode	✓
V3 Normal [17]	× landmarks missed	✓	✓	✓	—
V4 Sobel [18]	(✓) threshold	(✓)	(✓) preprocessing	✓	✓
V4.2 Gauss [19]	× no gain over V4	(✓)	✓	✓	—
V5 Hierarchical [20]	× segments unclear	✓	× nose bridge	✓	✓
V6 Kuwahara [21]	× at large radius	✓	✓	✓	✓

✓ satisfactory without tuning (✓) satisfactory after tuning × limitation observed

Five of the eight became candidates for the composite. None satisfied every criterion alone, and **facial readability** is the column that separated them.

Applying the Techniques

The same texture flaw defeats both detectors

V4.2 Gaussian prefiltered Sobel

V5 Hierarchical multi-cue

What the detectors produced

- Lines where there is no feature
- **Eyebrows and lip line barely register**
- V4.2: smoother, features still missing
- V5: brow and nose shadow become hard edges

Why

- Head rebuilt from photographs
- **Baked illumination** read as structure
- No threshold separates the two

Avatar Texture Manipulation

Fixing the input instead of the operator

Three Edits in Blender

- Baked illumination blurred out
- Skin tone unified
- Mouth saturation raised, lip line added

What It Recovered

- **Lip line and eyebrows** now detectable
- Gaussian prefilter no longer needed

Scope

- Applied to the **NPR avatar** only
- Photorealistic version unchanged

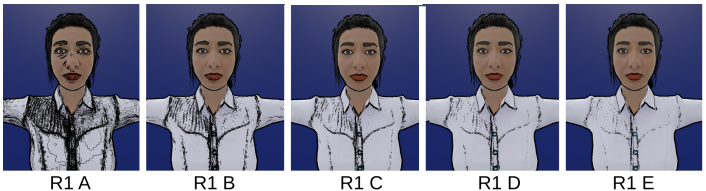
Default Avaturn

Preprocessed head texture

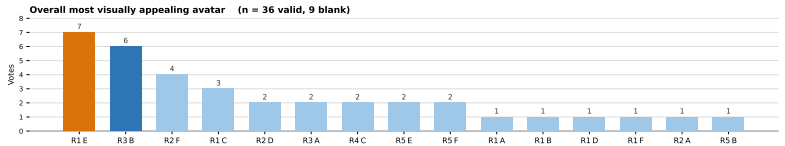
Parameter adjustment study

What participants ranked, and the closing appeal question

- **45 participants**, ranked by perceived trustworthiness
- One parameter group per round, then frozen for the rest
- Closing question: **which avatar looked most appealing overall**



One round as an example. R1 inner line threshold, 0.01 to 0.05.



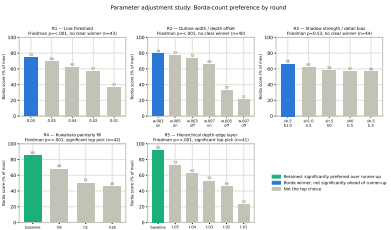
One vote per participant. **R1 E** first with seven, **R3 B** second with six.

Parameter adjustment study

Five rounds, and what each one settled

- Borda count, Friedman omnibus, Wilcoxon on the top pair [22]
- Three rounds found no separation, two did
- Ties broken by the **appeal vote**

	Parameter group	Winner	Significant?	Set to
R1	Inner line threshold	0.03	no, $p = .31$	**0.05** appeal
R2	Outline width, depth offset	0.003, on	no, $p = .82$	**0.003, on** Borda
R3	Shadow strength, detail bias	0.5 / 1.0	no, $p = .53$	**0.0** / **0.5** appeal
R4	Kuwahara hardness, sharpness	**none**	**yes,** $p = .003$	**off** baseline won
R5	Hierarchical depth threshold	**none**	**yes,** $p < .001$	**off** baseline won



Borda score per round. Green = significantly preferred, blue = top choice, not clearly ahead.

- **Kuwahara** and the **hierarchical layer** dropped
- Composite: **outline, X-Toon, Sobel**

RQ1 answer

How the techniques adapted across architectures, and what each platform imposed

Transfers with Retuning

- All eight techniques, all three platforms
- Same values, **visibly different** output per host
- Retuning needed everywhere

Needs a Substitute Signal

- Meta SDK: one fragment hook, no second pass
- No depth buffer, no normal buffer
- Hierarchical: camera-distance derivative, **not depth**
- Kuwahara: no Gaussian smoothing

The Dividing Line

- Not the technique's complexity
- New **parameter values**: always possible
- New **input signal**: only where the pipeline is open

Avaturn was chosen for the study: the only platform with open shader access, full blendshape animation and a photographic face.

RQ2 answer

Which combination balances abstraction against facial and expressive legibility

On the Raw Texture

- **None succeeded**
- **V3** normal edges: nothing on smooth skin
- **V4** Sobel: baked lighting read as structure
- **V5** hierarchical: brow and nose shadow become edges

After Preprocessing

- **V4** Sobel finds the **eyebrows and lip line**
- **V4.2** Gaussian no longer repays its 72 samples
- **Five of the eight** went forward as candidates

After the Parameter Study

- **V6** Kuwahara and **V5** hierarchical excluded by the ranking
- Composite: **V1 outline**, **V2.2 X-Toon**, **V4 Sobel**
- Plus the preprocessed head texture

The answer is a composite, not a technique. And it is not a technique alone either: without the texture preprocessing, none of the edge operators reached a readable face.

Main User Study

Within-subjects, video based, three conditions, one advisor task

Design

- 3 conditions $\times$ 3 scenarios: career, travel, learning
- 3×3 Latin square, **three versions**
- Scenario order fixed, conditions rotate

Paradigm after Canales et al. [9].

Measures per Block

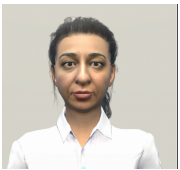
Likeability	Godspeed [23]	.94
Eeriness	Ho & MacDorman [3]	.87
Ability	Mayer [8], wording [9]	.96
Benevolence	Mayer [8], wording [9]	.95
Integrity	Mayer [8], wording [9]	.92

Sample

$n = 41$ of 52, 11 excluded · $M_{age} = 27.6$ ($SD = 5.4$) · 23 W, 17 M, 1 NB

C1 — Real

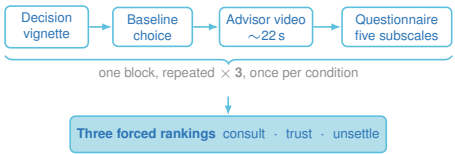
C2 — Photorealistic



C3 — NPR

Same person, same animation, same voice. **C2 vs C3** is the critical pair.

Study Procedure



Four Hypotheses, and All Five Subscales

What was predicted, and what the ratings did

4 Hypotheses & Statistical Tests

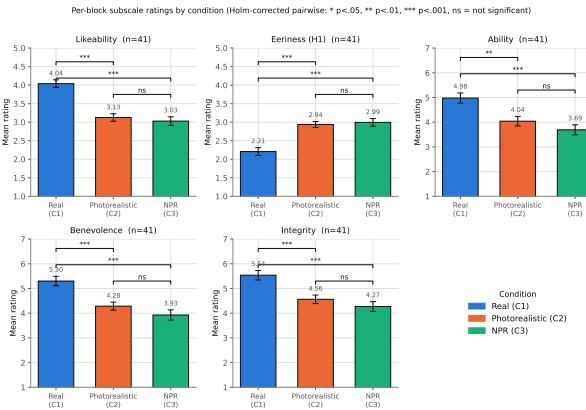
H1 NPR (C3) is rated *less eerie* than the photo-realistic avatar (C2).
Test: RM-ANOVA, Holm pairwise

H2 NPR (C3) does *not* cost trust relative to C2.
Test: Pairwise plus TOST equivalence, $d = 0.3$ bound

H3 Eeriness lowers trust, likeability raises it, within participant.
Test: Person-centred multilevel regression

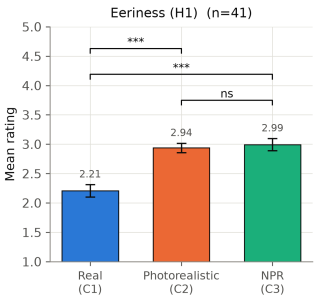
H4 NPR (C3) does *not* cost trust relative to the real speaker (C1).
Test: Same as H2, against C1

H2 and H4 predict a **null**. A non-significant *p* is not evidence of equivalence, hence the **TOST** bound.



The real speaker separates from both avatars on every subscale. The two avatars separate from each other on *none* of them.

H1 — Does NPR reduce eeriness?



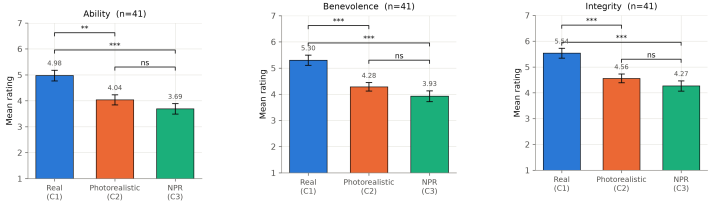
H1 NOT SUPPORTED

Eeriness	C1 vs C2	C1 vs C3	C2 vs C3
p	$< .001$	$< .001$	.66
g	0.98	0.98	0.07

Omnibus: $F(2, 80) = 20.39$, $p < .001$, $\eta_p^2 = .34$

- Eeriness rose from C1 to C2 to C3, the **opposite** of the prediction
- Real speaker differs from both avatars
- The two avatars **statistically indistinguishable**, $g = 0.07$
- Stylisation was seen, confirmed by the **final stylisation question**
- Seen, but not **less eerie**: two separate effects here

H2 — Does NPR cost trust against the photorealistic avatar?



ns between the two avatars on all three trust dimensions.

H2 SUPPORTED

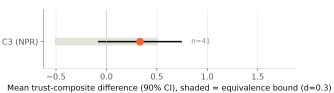
- **H2 holds as stated**: no trust dimension lower for C3
- **Equivalence unestablished**, the bound could not be ruled out
- Power near 0.20 at $n = 41$. Roughly **three times** the sample would settle it

Step 1. Pairwise, C2 vs C3

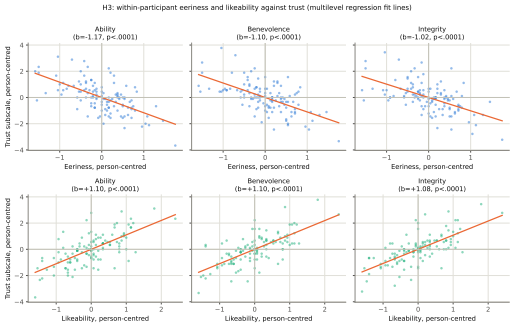
	p	g
Ability	.21	0.20
Benevolence	.18	0.19
Integrity	.27	0.16

Step 2. TOST, trust composite

	Diff.	Bound	TOST p
C2 vs C3	0.33	±0.51	.240



H3 — Eeriness, likeability and trust



H3 CONFIRMED

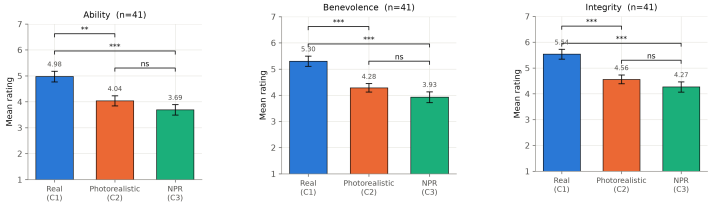
- **Within** participants, not across conditions
- Eeriness **lowers** trust, likeability **raises** it
- **One point for one point**, every dimension
- Confirmed in **both** directions
- Eeriness and likeability **move together**, $r = -.72$, hard to separate as independent variables

Predictor	Eeriness	Likeability
Ability	-1.17	$+1.10$
Benevolence	-1.10	$+1.10$
Integrity	-1.02	$+1.08$

Fixed-effect slopes, all $p < .001$

Takeaway. The route from appearance to trust is open. The composite never travelled it.

H4 — Does NPR cost trust against the real speaker?



The real speaker is **significantly above** the NPR avatar on all three.

Step 1. Pairwise, C1 vs C3

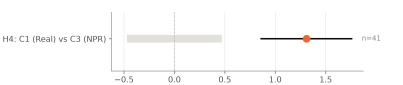
	p	g
Ability	$< .001$	0.75
Benevolence	$< .001$	0.79
Integrity	$< .001$	0.80

Step 2. TOST, trust composite

	Diff.	Bound	TOST p
C1 vs C3	1.31	± 0.47	.998

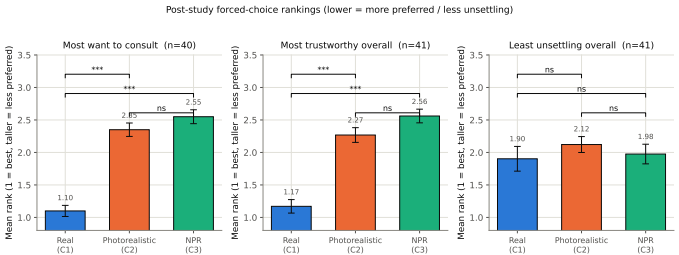
H4 NOT SUPPORTED

- All three subscales **significantly higher** for the real speaker
- Nearly **three times** the equivalence bound
- Not a marginal shortfall
- Two explanations stay open: the uncanny valley, or participants recognising the speaker



The behavioural measure

Forced ranking after all three videos, plus the checks on the null



Question	Friedman	C1 vs C2/C3	C2 vs C3
Most want to consult	$\chi^2_F(2) = 49.40, p < .001$	$p < .001$	$p = .21$
Most trustworthy	$\chi^2_F(2) = 44.05, p < .001$	$p < .001$	$p = .07$
Least unsettling	$\chi^2_F(2) = 1.02, p = .60$	$p = .97$	$p = .97$

Canales et al. [9] found ratings and behaviour *diverge*. Here they **converged**. C3 attracted no more behavioural endorsement than C2.

Robustness of the Null

- **Order**: video position affected no subscale, closest $p = .054$
- **Suspicion**: 24 of 41 guessed the topic, and rerunning H1 and H2 on the other 17 changed nothing
- **Secondary**: agreement with the recommendation did not differ, $p = .13$
- **Disposition**: general trustingness did not predict the composite, $p = .52$

Study version groups differed on likeability and eeriness but on no trust subscale, and that comparison is between subjects.

RQ3 answer

What stylisation did to eeriness and trustworthiness, and how far it reached

What Did Not Happen

- Eeriness **did not fall**, it rose slightly
- The two avatars **indistinguishable** on all five subscales
- No behavioural advantage either

What Did Happen

- **No trust penalty** detected for stylisation
- Within a person, eeriness **lowers** trust and likeability **raises** it
- About **one point for one point**, on all three trust dimensions
- The route from appearance to trust is real, the composite just never used it

The Dividing Line

- The gap is **not between render styles**
- C2 vs C3: 0.33 points, inside the noise
- C1 vs C3: 1.31 **points**, nearly three times the bound

Stated carefully: H2 is *supported*, but the two avatars are **indistinguishable rather than demonstrably equivalent**. Realised power near 0.20 at $n = 41$, so the honest claim is that no penalty was detected, not that none exists.

Discussion

Why the two avatars scored the same

Two mismatches inside the composite

The real speaker sits far above both avatars, and the two avatars looked different to participants yet scored identically. H3 showed the route from appearance to trust was **open**. The composite changed what people saw **without travelling it**.

1. Material versus Shape

- The composite stylises the **material**, leaving the **shape** untouched
- Zell et al.: material is what drives eeriness [24]
- But eeriness **rises** when material and shape disagree in stylisation level
- C3 is exactly that case: a stylised surface on a photogrammetric face
- Stronger stylisation would only **deepen** the disagreement

2. Appearance versus Performance

- Both avatars carry the **same** recorded voice, gaze and lip animation
- C3 sets a **human performance** against a visibly abstracted face
- Cihodaru-Ştefanache & Podina: uncanniness arises from **inconsistent** levels of anthropomorphism [25]
- Kätsyri et al.: face and voice realism varied independently, and **mismatched pairs were rated eerier** [26]
- Beyond the composite's reach: a shading operator changes appearance alone

Both point the same way. The abstraction may have **removed one source of eeriness and introduced another**, and both widen as the abstraction is pushed further.

What this design could not show

Two reasons the effect may exist and still not appear here

The Comparison Set

- A real person anchors the top of the scale
- The only judgement left is **how far below**
- Both avatars recognisably CG, so a **rendering** difference competes with a **categorical** one
- C3 followed C1 directly in two versions of three, C2 never did

The Video Setting

- Flat screen, prerecorded, no interaction, no shared space
- Zibrek et al.: render style significantly moved **presence** in VR [27]
- Too weak in a browser window may register once the avatar **shares your room**

Neither is a defect in the analysis. Both say the same thing: this design tests stylisation **under conditions that work against it**.

Limitations

Six, and each one bounds a different claim

Session Length

- 10 to 15 minutes per study
- Every measure completed **three times**
- Capped the measures and the breadth of the design

One Configuration

- One composite, one avatar, one platform, three scenarios
- Characterises **this configuration**, not NPR as a category

Parameter Study Scope

- Only the **highest-impact** parameter groups tested
- The rest fixed by inspection in development
- Highest **among those tested**, not necessarily the best available

Sample Size

- Deadline set the collection window, in the semester break
- $n = 41$ **detects** a difference
- **Not** the far larger sample equivalence demands

Counterbalancing

- Three conditions need **six** orders
- Six orders need twice the sample
- Three were used instead

Version Group Differences

- Version groups differed on **eeriness and likeability**, which is what **H1 rests on**
- Version sets both *who* and *what order*, and the two cannot be separated
- This **qualifies the H1 null**

Conclusion

What was built, what it showed, and what it means

What This Thesis Built

- NPR stylisation for **Mixamo, the Meta SDK and Avaturn**
- Eight techniques in sequence, each answering a limitation of the one before
- Narrowed to **three** by visual assessment and a parameter adjustment study

What the Study Showed

- The composite **did not reduce eeriness** against the photorealistic baseline
- It produced **no observed trust penalty** either
- The evidence does not indicate stylisation reduced trust
- It is **not sufficient** to establish that the two are equivalent

What It Means

- A developer choosing between the two styles need not assume **an inherent disadvantage**
- The gap that matters is **not between render styles**
- It is the distance to the **real human**, and neither avatar closed any part of it

Reaching the credibility of an actual human is a harder challenge than selecting a rendering style.
It is also the problem that decides whether virtual advisors become genuinely trusted participants rather than a passing novelty.

Future work

Five steps, in the order the thesis sets them out

1. Larger Replication

- Repeat C2 versus C3 with roughly **120 participants**
- Settles the equivalence question either way

2. Stylise the Geometry

- Shift proportion and feature scale **alongside the shading**
- Tests the material-versus-shape mismatch

3. Drop the Real Speaker

- Removes the human **anchor**
- Shows whether the two avatars separate with no human in view

4. Sweep Abstraction

- Replace the single point with a **curve**
- Can reveal an optimum between full realism and full abstraction

5. Restore Interaction

- Reciprocal behaviour and shared space, in **VR**
- Across multiple avatar identities and platforms

One smaller fix: **state at the outset that the videos were not generated by AI**. Nothing in the present materials ruled out that belief, and it may have lowered trust for both avatar conditions.

Thank you

Questions

Eight NPR techniques, three avatar platforms, two studies, 86 participants.

Stylisation was clearly seen, cost no measurable trust,
and did not close the gap to a real human.

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Media credits

Video, slide 2

Mark Zuckerberg: First Interview in the Metaverse, Lex Fridman Podcast #398. YouTube.
youtube.com/watch?v=MVYrJJNdrEg

Avaturn: Realistic 3D Avatars for Gaming from Photo. YouTube.
youtube.com/watch?v=rSb0lo92_R8

How Apple Vision Pro's New Spatial Persona Changes Everything. YouTube, 11 April 2024.
youtube.com/watch?v=H3PB2zA6Q-c

Excerpts shown for scholarly illustration under the German right of quotation, § 51 UrhG.

Figures

Slide 3, uncanny valley curve: Mori, reference [1].

Slide 4, android photograph: reference [2].

Slide 5, render styles: reference [3].

Slides 4 to 6: every figure is reproduced from the study cited in the same panel. The number on the title line covers the figure as well as the claim, references [1] to [13].

Slides 10, 11, 14, 15, 17, 18 and 20, and all backup slides: own work. Renders from the Unity project, plots from the study data.

Avatars: Avaturn, Meta Avatars SDK and Mixamo, used under their respective terms.

Logo: University of Konstanz.

Backup slides

Backup — How the evidence developed

Fifty years of it, and it still points both ways

Period	What changed	The evidence	Points towards
1970 to 2012	The valley is real, and it reaches avatars	Mori proposes it [1], MacDorman & Ishiguro measure it [2], Dill et al. confirm it for CG film and game characters [4]	valley confirmed
2012	Render style moves *trust*, in the same shape	McDonnell et al.: eleven styles on two actors. Appeal, reassurance and trustworthiness all valley-shaped, and the cartoon styles sat in the top trust group [6]	**abstraction**
2015	What drives it is the material, and consistency	Zell et al.: material governs eeriness, and a mismatch between material and shape makes it worse [24]	depends on the pairing
2024 to 2025	Abstraction looks promising	Canales et al.: self-report equal, semi-realistic won the behavioural choice [9]. Dubosc et al.: stylised significantly less eerie [10]	**abstraction**
2021 to 2026	But realism keeps winning too	Seymour et al.: photorealistic beat a caricature on trust [28]. Gao et al.: real speaker higher on all three dimensions [11]. Tao et al.: high realism more trustworthy [12]. Alipour et al.: inconsistent across 53 studies [13]	**realism**

The last two rows are the reason this thesis exists. The evidence has not converged, and nobody has tested one avatar under both treatments against the real person it was built from.

Backup — V1 — Silhouette outline

Inverted hull, drawn from geometry alone

Mixamo (Jody)

Meta SDK

Avaturn

Backup — V2 — Toon shading

The diffuse term quantised into bands

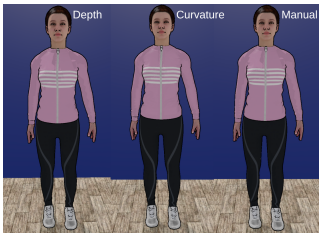
Mixamo (Jody)

Meta SDK

Avaturn

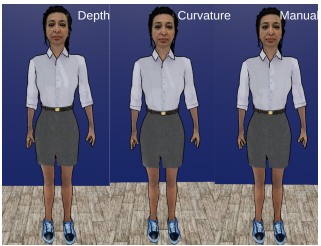
Backup — V2.2 — X-Toon

A 2D ramp: lighting on one axis, abstraction on the other



Mixamo (Jody)

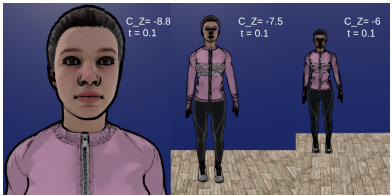
Meta SDK



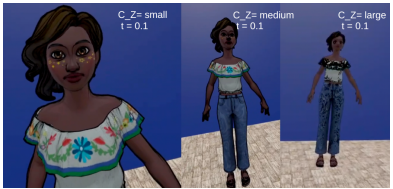
Avaturn

Backup — V3 — Screen-space normal edges

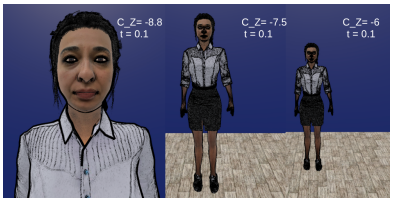
Normal derivatives. Creases yes, colour boundaries no



Mixamo (Jody)



Meta SDK



Avaturn

Backup — V4 — Sobel

Gradient on the colour buffer, at rising thresholds

Mixamo (Jody)

Meta SDK

Avaturn

Backup — V4.2 — Gaussian prefiltered Sobel

Smoothing before differentiation

Mixamo (Jody)

Meta SDK

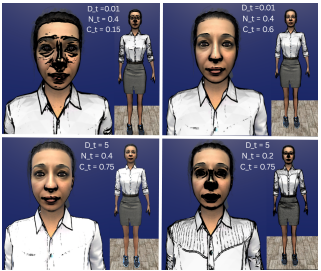
Avaturn

Backup — V5 — Hierarchical multi-cue

Depth, normal and colour layers merged

Mixamo (Jody)

Meta SDK



Avaturn

Backup — V6 — Kuwahara

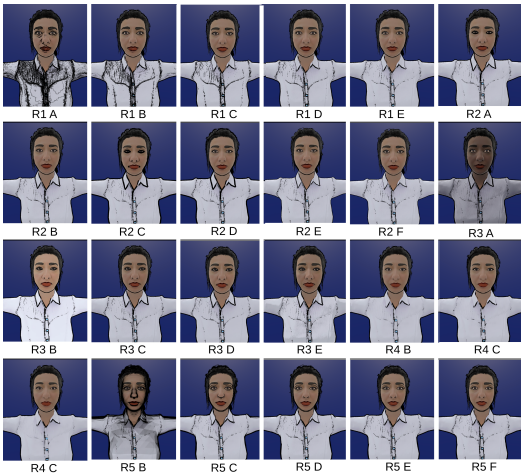
Anisotropic painterly filtering of the surface

Mixamo (Jody)

Meta SDK

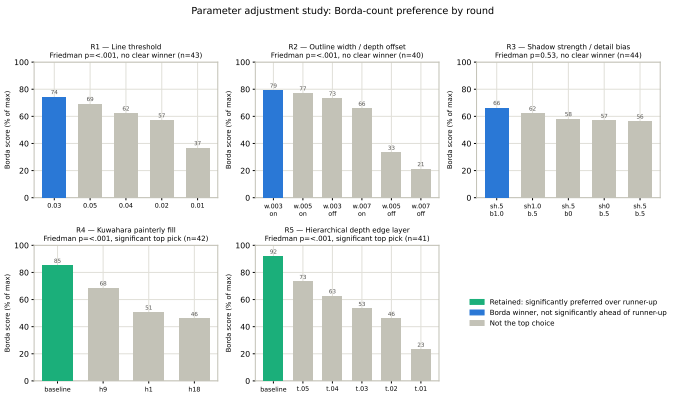
Avaturn

Backup — All variants shown in the parameter adjustment study



Backup — Borda score by round

Preference score as a percentage of the maximum possible



Green marks the baseline kept after a significant preference in R4 and R5. Blue marks the Borda winner in R1 to R3, where the top choice was not significantly ahead of its runner-up.

Backup — The three scenarios

Order held constant, conditions rotated by version

	Scenario	The two options	Advisor picks	Primary trust dimension
Video 1	Career choice	Job A generalist, lower pay Job B specialist, higher pay	**Job A**	Ability
Video 2	Travel route	Route A direct, 10 min spare Route B one transfer, 40 min spare	**Route B**	Benevolence, Ability
Video 3	Learning resource	Option A course, EUR 350, certificate Option B free videos, no certificate	**Option A**	Integrity

Counterbalancing

Version	Video 1	Video 2	Video 3
A	C2	C1	C3
B	C3	C2	C1
C	C1	C3	C2

Design Rules

- Recommended option alternates **A, B, A**
- Both options given equal detail
- Advisor states only its recommendation and reasoning
- Baseline recorded **before** the advisor speaks

Backup — Vignette 1, Career choice

Shown on screen before the video

You recently completed your master's degree and have received two job offers. Both start in one month. You have five days to decide.

Job A is a generalist role at a smaller company. You would define your own work area, your contributions would be directly visible to the team, and the skills you build would transfer across industries. The starting salary is below the industry average for your field.

Job B is a specialist role at a large, well-known company. The employer name carries weight on a CV and the role is in a high-demand field. However, the position involves deep specialisation within a large team, with limited individual visibility. The starting salary is above the industry average for your field.

Advisor, spoken in the video

"I'd go with Job A. Yes, the salary is lower right now, but the generalist role gives you a range of experience that is genuinely hard to get later. Salary gaps can be closed in the future. But once you specialise, switching back is costly. Take the pivot while you can."

Backup — Vignette 2, Travel route

Shown on screen before the video

You have just arrived by train in a city you have never visited. You need to reach your hotel, where colleagues are expecting you in one hour.

Route A is a direct route on a single line with no transfers. It is straightforward to navigate in an unfamiliar city. If the journey runs on time, you would arrive approximately 10 minutes before your meeting begins.

Route B involves one transfer at a mid-size station you have not used before. If the connection runs on time, you would arrive approximately 40 minutes before your meeting begins. However, the transfer station is unfamiliar and a missed connection could leave you with very little time.

Advisor, spoken in the video

"I'd take Route B. Yes, it has a transfer at a station you have never used. But it gets you there forty minutes early. If something goes wrong, you still have enough time to recover. Route A is direct but leaves only ten minutes. That may not be enough. Take the buffer."

Backup — Vignette 3, Learning resource

Shown on screen before the video

You want to develop a professional skill that would strengthen your CV and improve your career prospects. You have a few weeks available and you are a student with a limited budget.

Option A is a structured online course from a well-known learning platform. It follows a clear curriculum and awards a recognised certificate you can add to your CV. It costs EUR 350.

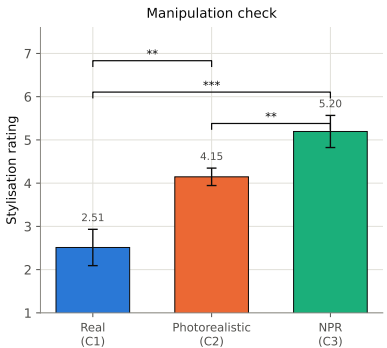
Option B is a video tutorial series by a practitioner with a strong reputation. It covers the same material at no cost and you set your own pace. However, there is no certificate and no structured curriculum.

Advisor, spoken in the video

"I would take Option A. Yes, 350 euros is a real cost. But the certificate is what matters here. It is not just about what you learn, it is about what you can show. A recruiter cannot verify a free tutorial series. That is worth it."

Backup — Manipulation check

Did participants actually see a difference?



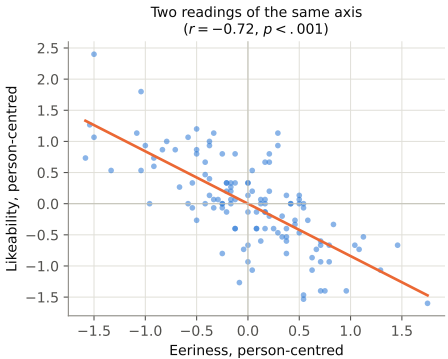
Part H asked how stylised each advisor looked, from 1 (completely realistic) to 7 (very stylised).

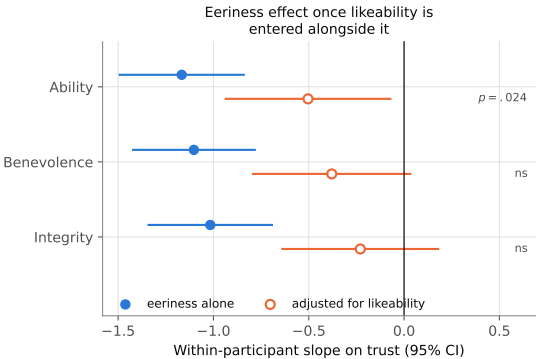
Omnibus	$F(2, 80) = 15.45$, $p < .001$, $\eta_p^2 = .28$
C1 vs C2	$p = .001$, $g = 0.87$
C1 vs C3	$p < .001$, $g = 1.23$
C2 vs C3	$p = .004$, $g = 0.55$

Ratings rise monotonically from Real to Photorealistic to NPR. All three conditions separate.

Takeaway. The manipulation worked. No null result below can be blamed on participants not noticing the rendering.

Backup — Eeriness against likeability





Within participants the two correlate at $r = -.72$, sharing 52% of their variance. Entering both in one model absorbs most of the eeriness effect: it survives on ability ($b = -0.50$, $p = .024$) but not benevolence ($p = .075$) or integrity ($p = .273$), while likeability holds on all three.

Backup — Full statistics table

Subscale	Omnibus	C1 vs C2	C1 vs C3	C2 vs C3
Likeability	$F(2, 80) = 26.85$, $p < .001$, $\eta_p^2 = .40$	$p < .001$, $g = 0.92$	$p < .001$, $g = 1.02$	$p = .54$, $g = 0.08$
Eeriness	$F(2, 80) = 20.39$, $p < .001$, $\eta_p^2 = .34$	$p < .001$, $g = 0.98$	$p < .001$, $g = 0.98$	$p = .66$, $g = 0.07$
Ability	$F(2, 80) = 10.91$, $p < .001$, $\eta_p^2 = .21$	$p = .004$, $g = 0.53$	$p < .001$, $g = 0.75$	$p = .21$, $g = 0.20$
Benevolence	$F(2, 80) = 14.31$, $p < .001$, $\eta_p^2 = .26$	$p < .001$, $g = 0.59$	$p < .001$, $g = 0.79$	$p = .18$, $g = 0.19$
Integrity	$F(2, 80) = 12.47$, $p < .001$, $\eta_p^2 = .24$	$p = .001$, $g = 0.66$	$p < .001$, $g = 0.80$	$p = .27$, $g = 0.16$

Order and version checks

	Position		Study version	
Subscale	$F(2, 80)$	p	$F(2, 38)$	p
Likeability	3.04	.054	**7.07**	**.002**
Eeriness	2.47	.091	**5.57**	**.008**
Ability	1.32	.274	1.23	.304
Benevolence	2.10	.130	1.26	.296
Integrity	1.72	.186	2.04	.144

Intraclass correlations, H3 models

Eeriness models: .48 to .57.
Likeability models: .57 to .63.

Half or more of the variance in each trust subscale sits *between* participants rather than within them, which is consistent with large individual differences in general trustingness.